

Mathematical modelling and control of CAS

Sigurd R. Brekk

August 7, 2025

1 Introduction

Dynamical model of CAS system.

Formula for flow vs speed: $Q = K_1 \cdot H \cdot n$, where n is the impeller speed in RPM, and Q is m^3/h . For the SR2000 pump the area of operation is between 40-130 rpm. If we linearize around the rated point and switch to units m^3/s , we get:

$$\frac{20000}{60^2} = K_1 \cdot 130 \implies K_1 \approx 0.042735.$$

The flow out of the pen is designed so we have an exchange rate of 35 minute for the entire volume of the pen which is $20000m^3$. If we switch units to m^3/h and linearize around 0.06m, we get:

$$K_{qo} = \frac{20000m^3}{35 \cdot (60s)} \approx 9.5238m^3/s \quad (1)$$

$$q_o = \frac{K_{qo}}{0.06} \cdot x_1 \quad (2)$$

where x_1 is the height difference inside-outside the pen. C is the capacitance of the tank

$$C = \frac{\partial V}{\partial h} = \pi r^2 = \pi \cdot 25.25m^2 \quad (3)$$

$$\approx 1288.25m^2 \quad (4)$$

$$C\dot{x}_1 = q_i - q_o \quad (5)$$

$$= K_1 \cdot u_1 - \frac{K_{qo}}{0.06} x_1 \quad (6)$$

$$\dot{x}_1 = -\frac{K_{qo}}{0.06 \cdot C} x_1 + \frac{K_1}{C} u_1 \quad (7)$$

1.1 Oxygen model

$$\dot{a} = c_{in} \cdot r_{in} - c_{out} \cdot r_{out} \quad (8)$$

Where a is the amount of oxygen in the system, $c_{in} \left[\frac{kg}{m^3} \right]$ is the concentration going in, c_{out} is the concentration going out. $r_{in} \left[\frac{m^3}{h} \right]$ is the rate of the amount going in and r_{out} is the rate of oxygen going out. Substituting this with the state variables the equation becomes when linearized around 0.06cm

$$\dot{x}_2 = +K_2(u_1 + u_2 + u_3) + x_3 - K_4 - K_{qo} \cdot \frac{x_2}{20000} \quad (9)$$

where $F(s)$ is the transfer function between We also have an additional contribution to x_2 from the oxygenation system, which is on the from

$$\frac{X_2}{U_4} = \frac{K_3}{1 + T_1 s} - K_{qo} \cdot \frac{x_2}{20000}$$

Where K_2 is the amount of oxygen inside the sea water (concentration times flow). u_1, u_2, u_3 are the control signals to the inlet pump (in RPM). u_4, u_5, u_6 is the control signal to the control valve for the oxygenation system. $G_1(s)$ is the transfer function from the control signal to the valve to the outout flow K_4 is the assumed constant consumption of oxygen from the fish.

$$G_1(s) = \frac{K_5}{1 + T_1 s} \quad (10)$$

K_5

The constant K_2 is relating the pump speed to rate of oxygen that is inserted in the pen $\frac{g}{h}$.

Normal O_2 levels in sea water is $\frac{7g}{m^3}$.

From equation 7 we have the relationship between RPM and oxygen into the CAS (with assumption of constant oxygen concentration of $\frac{7g}{m^3}$).

$$\frac{x_2}{u_1} = K_2 = 7 \frac{g}{m^3} \cdot K_1 \approx 0.299145 \frac{g}{s} \quad (11)$$

Maximum o_2 injection pr. inlet branch is specified to be $104.2kg/h$ ($\approx 29g/s$) If we assume a linear relationship between the control signal and outlet flow, where the maximum oxygen flow is $\approx 29g/s$ pr. specification, we get:

$$\frac{x_2}{u_4} = K_5 = \frac{1}{100} \cdot 29 \frac{g}{s} = 2.9 \frac{g}{s} \quad (12)$$

where u_4 is a control signal between 0-100%.

The time constant for the Burkert 2865 valve is given to $< 20ms$. Whit the possibility to introduce a ramp delay between 0..10s

Equation 10 then becomes

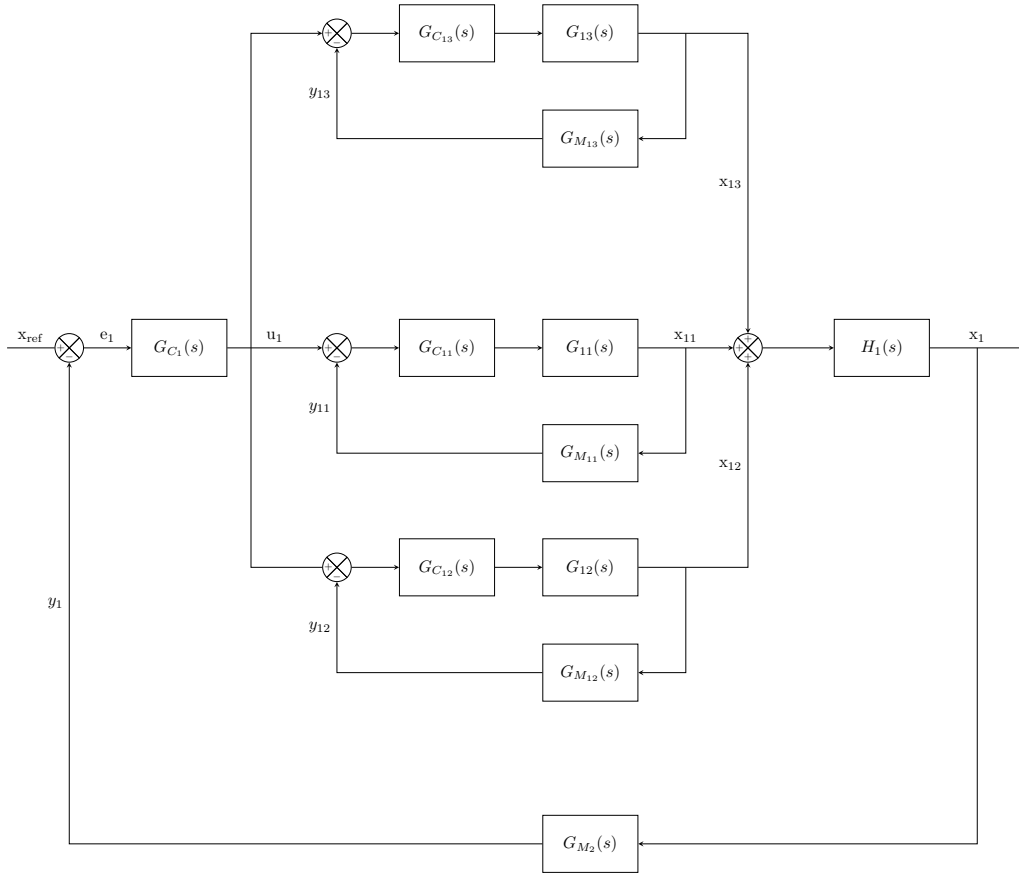
$$G_1(s) = \frac{2.9}{1 + 0.02s} \quad (13)$$

The average consumption of fish in this model is $500mgO_2/kg h$. If we switch units to seconds and gram, and assume a total of 1 000 000kg salmon we get the value of K_4 to be

$$500 \cdot \frac{1000000}{60^2 \cdot 1000} \frac{gO_2}{s} \approx 139gO_2/s$$

2 Block diagram of oxygen system

G_{C_1} - oxygen controller
 $G_{C_{11}}, G_{C_{12}}, G_{C_{13}}$ - flow controller
 H_1 - plant
 $G_{M_{11}}, G_{M_{12}}, G_{M_{13}}$ - measurement
 G_{11}, G_{12}, G_{13} - flow valve



3 State space description of the system

$$\dot{\mathbf{x}}(\mathbf{t}) = \mathbf{A}\dot{\mathbf{x}}(\mathbf{t}) + \mathbf{B}\mathbf{u}(\mathbf{t}) \quad (14)$$

$$\mathbf{y}(\mathbf{t}) = \mathbf{C}\mathbf{x}(\mathbf{t}) \quad (15)$$

linearize around inner head at 6cm

x_1 = inner water height difference	u_1 = pump 1 speed [rpm]
x_2 = amount of oxygen inside the pen	u_2 = pump 2 speed [rpm]
x_3 = O2 flow injector number 1 $\left[\frac{g}{s}\right]$	u_3 = pump 2 speed [rpm]
x_4 = O2 flow injector number 2 $\left[\frac{g}{s}\right]$	u_4 = control signal to O2 injector valve 1 [0-100%]
x_5 = O2 flow injector number 3 $\left[\frac{g}{s}\right]$	u_5 = control signal to O2 injector valve 2 [0-100%]
	u_6 = control signal to O2 injector valve 3 [0 – 100%]
	u_7 = Oxygen consumption of fish $\frac{g(O_2)}{s}$

Constants:

ρ = 1025 sea water density	K_1 = constant relating RPM to g/m^3 of oxygen
$C = \frac{\partial V}{\partial h} = \pi \cdot r^2 \approx 1886$ Capacity of tank	$K_1 = 6512.5$ gain of control signal to O2 massflow
$T_1 = 2$ time constant of O2 control valve	$V = 20000$ volume of tank
$K_{qo} = 1.749985 \cdot 10^{-6}$ Resistance of outlet	

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \end{bmatrix} = \begin{bmatrix} -\frac{K_{qo}}{0.06 \cdot C} & 0 & 0 & 0 & 0 \\ 0 & -\frac{K_{qo}}{V} & 1 & 1 & 1 \\ 0 & 0 & -\frac{1}{T} & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{T} & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{T} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} + \begin{bmatrix} \frac{K_1}{C} & \frac{K_1}{C} & \frac{K_1}{C} & 0 & 0 & 0 & 0 \\ K_2 & K_2 & K_2 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & K/T & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & K/T & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & K/T & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \end{bmatrix} \quad (16)$$

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{V} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \quad (17)$$

4 ZOH discretization of system

Z-transforms and difference equations

$$\frac{Y_1(z)}{U_1(z)} = \frac{3.29694276e - 06}{z - 0.98775421} \quad (18)$$

$$\frac{Y_2(z)}{U_1(z)} = \frac{1.49569088e - 06}{z - 0.99995238} \quad (19)$$

$$\frac{Y_2(z)}{U_4(z)} = \frac{(3.55844420e - 06)z + 3.49957382e - 06}{z^2 - 1.95118181z + 0.95118413} \quad (20)$$

$$\frac{Y_2(z)}{U_7(z)} = \frac{-4.99988095e - 06}{z - 0.99995238} \quad (21)$$

$$\frac{Y_2(z)}{X_3(z)} = \frac{0.09999762}{z - 0.99995238} \quad (22)$$

$$\frac{X_3(z)}{U_4(z)} = \frac{28.74941831}{z - 0.006738} \quad (23)$$

$$(24)$$

$$y_{1_k} = 0.98775421 \cdot y_{1_{k-1}} + 3.29694276 \times 10^{-6} \cdot u_{1_{k-1}}$$

$$y_{2_k} = 0.99995238 \cdot y_{1_{k-1}} + 1.49569088 \times 10^{-6} \cdot u_{1_{k-1}}$$

$$y_{2_k} = 1.95118181 \cdot y_{2_{k-1}} - 0.95118413 \cdot y_{2_{k-2}} + 3.55844420 \times 10^{-6} \cdot u_{4_{k-1}} + 3.49957382 \times 10^{-6} \cdot u_{4_{k-2}}$$

$$y_{2_k} = 0.99995238 \cdot y_{2_{k-1}} - 0.00069443 \cdot u_{7_{k-1}}$$

$$y_{2_k} = 0.99995238 \cdot y_{2_{k-1}} + 0.09999762 \cdot x_{3_{k-1}}$$

$$x_{3_k} = 0.006738 \cdot x_{3_{k-1}} + 28.74941831 \cdot u_{4_{k-1}}$$

Final value theorem:

$$x[\infty] = \lim_{x \rightarrow 1} (z - 1)X(z) \quad (25)$$

Z-transform of a step function is $\frac{z}{z - 1}$

5 Control theory method for regulation of CAS

6 Results